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BCACACN 202

**Second Semester B.C.A. Degree Examination, June/July 2024
(NEP – 2020) (2021 – 22 Batch Onwards)
DATA STRUCTURES USING C (DSCC)**

Time : 2 Hours

Max. Marks : 60

Note : Answer *any six* questions from Part – A and *one full* question from *each* Unit in Part – B.

PART – A

(6×2=12)

1. a) Define linear and non-linear data structure. Give an example.
- b) What is sparse matrix ? Give an example.
- c) Define searching. Name any 2 types of searching techniques.
- d) Write an advantage of linked list over arrays.
- e) What is stack ? Name the basic operations performed on stack.
- f) What is garbage collection ?
- g) Name the 2 ways of representing trees in memory.
- h) Define the terms with respect to graph.
 - i) adjacent node.
 - ii) directed edge.

PART – B

UNIT – I

(4×12=48)

2. a) What do you mean by traversing a linear array ? Write an algorithm to traverse a linear array. **(4+4+4)**
- b) Write an algorithm to generate 'N' Fibonacci numbers using Recursion.
- c) Explain bubble sort with an example.
3. a) Explain insertion sort with an example. **(4+4+4)**
- b) Write an algorithm for selection sort.
- c) Explain the representation of two dimensional array in memory.

P.T.O.



UNIT – II

4. a) What is linear search ? Write an algorithm for linear search. (4+4+4)
b) What is circular linked list ? Explain.
c) Write an algorithm to insert an item into the beginning of the linked list.
5. a) Explain binary search with an example. (4+4+4)
b) Write an algorithm to delete a given node from the linked list.
c) Write a note on memory allocation and deallocation functions.

UNIT – III

6. a) Write an algorithm for PUSH and POP operations using arrays. (4+4+4)
b) Write an algorithm to convert infix expression into postfix expression.
c) What is priority queue ? Explain.
7. a) Write an algorithm to delete an item from stack using linked list. (4+4+4)
b) Evaluate the following postfix expression showing the stack status.
P : 3, 5, +, 6, 4, -, *, 4, 1, -, 2, ^, +.
c) Write an algorithm to insert an element into a queue using linked list.

UNIT – IV

8. a) Write a note on : (4+4+4)
i) Complete binary tree.
ii) Extended binary tree.
b) Define the terms with respect to tree.
i) Root node.
ii) Terminal nodes.
iii) Degree of a node.
iv) Depth.
c) Write an algorithm for Depth First Search (DFS) for a graph.
9. a) Draw the binary tree for the following inorder and preorder traversal. (6+6)
Inorder : E, A, C, K, F, H, D, B, G
Preorder : F, A, E, K, C, D, H, G, B
And also give its postorder traversal.
b) Explain linked representation of binary tree with an example.
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BCACACN 202

**Second Semester B.C.A. Degree Examination, July/August 2023
(NEP 2020) (2021 – 22 Batch Onwards)
DATA STRUCTURES USING C (DSCC)**

Time : 2 Hours

Max. Marks : 60

Note : Answer *any six* questions from Part – A and *one full* question from *each* Unit in Part – B.

PART – A

1. a) Define linear data structure. Give an example. (6×2=12)
b) Consider the linear array AA(5:50). Find the number of elements in an array.
c) What is sparse matrix ? Give example.
d) Write one advantage and disadvantage of binary search.
e) What is Linked List ? List the different types.
f) Write the algorithm for POP operation in Stack using array.
g) Define the terms with respect to tree :
 i) leaf node
 ii) depth of a tree.
h) Define adjacency matrix.

PART – B

Unit – I

2. a) Write the classification of data structure and briefly explain it.
b) Write an algorithm for Bubble sort.
c) Explain the selection logic with example. (4+4+4)
3. a) Explain the representation of two dimensional array in memory.
b) Trace the following list of numbers using selection sort technique :
 77, 33, 44, 11, 88, 22, 66, 55. (6+6)

P.T.O.

**Unit – II**

4. a) What is Linear Search ? Write the algorithm for linear search.
b) Explain any two dynamic memory allocation functions used in 'C' with syntax and example.
c) Write the algorithm to insert a node at the beginning of the linked list. (4+4+4)
5. a) Explain binary search with an example.
b) Write an algorithm to delete a node following a given node of a linked list.
c) Explain the representation of linked list in memory. (4+4+4)

Unit – III

6. a) Write an algorithm to evaluate postfix expression using stack.
b) Write an algorithm to perform insertion operation on queue using array.
c) Convert the infix expression to postfix : $a * b - c ^ d + e / f$. (4+4+4)
7. a) Using Stack convert the given infix expression to postfix.
 $(A + B) * (C - D) / E * F$
b) Write an algorithm for PUSH operation onto a Stack using Linked List.
c) Evaluate the following Postfix expression showing the Stack status.
 $P : 5, 6, 2, +, *, 1, 4, 2, /, ^, *$ (4+4+4)

Unit – IV

8. a) Draw the Binary Search Tree for the following numbers and Traverse it in Preorder, Inorder and Postorder :
60, 40, 100, 80, 45, 35, 120, 105
b) Explain the method of representing the graphs using sequential method with an example. (6+6)
9. a) Draw the binary tree for the expressions given below. Also traverse in preorder, postorder and inorder :
 $A + B - C ^ D / E * F$
b) Write an algorithm for breadth first search (BFS) for a graph. (6+6)
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BCACACS 201

**Second Semester B.C.A. Degree Examination, June/July 2025
(SEP) (2024 – 2025 Batch Onwards)
DATA STRUCTURES**

Time : 3 Hours

Max. Marks : 80

Note : Answer *any ten* questions from Part – A and *any one full* question from *each* Unit in Part – B.

PART – A

(10×2=20)

1. a) Define linear data structure. Give an example.
- b) Define sorting. Why it is necessary ?
- c) Consider the linear array LA (1 : 10). Find the number of elements in an array.
- d) What do you mean by dynamic memory allocation ? Write any 2 functions used in 'C' for dynamic memory allocation.
- e) What is two-way list ? Give its diagram.
- f) Write the advantages of linked list over arrays.
- g) Write an algorithm for PUSH operation using arrays.
- h) Define dequeue. What are its types ?
 - i) Write an algorithm to find factorial of a number N using recursion.
 - j) Define complete binary tree. Give an example.
- k) What is Graph ? Give an example.
- l) Define the terms with respect to graph.
 - i) Adjacent node
 - ii) Directed edge.

P.T.O.





PART – B

Unit – I

2. a) Write the classification of data structure and explain.
b) Trace the following numbers using bubble sort for the following data :
65, 25, 10, 80, 25, 11, 45, 20.
c) Write an algorithm to inserting an element into a linear array. (5+5+5)
3. a) Explain the various algorithmic notation.
b) Write an algorithm for selection sort.
c) Explain the memory representation of one dimensional array. (5+5+5)

Unit – II

4. a) Explain the memory representation of linked list.
b) Write an algorithm for binary search.
c) Write an algorithm to insert an item at the beginning of a linked list. (6+5+4)
5. a) What is linked list ? Explain the different types.
b) What is linear search ? Write the algorithm for linear search.
c) Write an algorithm to delete a node following a given node of a linked list. (6+5+4)

Unit – III

6. a) Write an algorithm to evaluate postfix expression using stack.
b) Convert the following infix expression into postfix expression using stack status.
Q : $((A + B) * D) ^ (E - F)$.
c) Write an algorithm to perform insertion operation on circular queue using array. (5+5+5)



7. a) Write an algorithm to convert infix to postfix expression using stack.
b) Evaluate the following postfix expression showing the stack status.
P : 3, 5, +, 6, 4, −, *, 4, 1, −, 2, ^, +.
c) Write an algorithm to perform insertion operation on queue using linked list. (5+5+5)

Unit – IV

8. a) Draw the binary search tree for the following data and traverse in all 3 methods
66, 26, 22, 34, 47, 79, 48, 32, 78.
b) Explain the method of representing the graphs using sequential method with an example.
c) Write an algorithm for Depth First Search (DFS) for a graph. (5+5+5)
9. a) With an example, explain sequential representation of binary tree.
b) Draw the binary tree for the expressions given below. Also traverse in preorder, postorder and inorder.
 $A * B + (C \wedge D/E/F)$.
c) What is adjacency matrix and path matrix, explain with an example. (5+5+5)
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